

Maximal Semantic Clique Enumeration in Vector-Attributed Graphs

Tianyi Gao

Beijing Institute of Technology
Beijing, China
tygao_bit@163.com

Hongchao Qin

Beijing Institute of Technology
Beijing, China
qhc_neu@gmail.com

ABSTRACT

Vector-attributed graphs encode two complementary forms of group cohesion: edges record structural relations, while vertex vectors describe semantic compatibility. Maximal clique enumeration captures the former but provides no criterion for the latter. We study maximal semantic clique (MSC) enumeration, which retains complete connectivity and requires the average pairwise vector similarity to reach a threshold. Unlike an every-pair threshold, this aggregate criterion tolerates a small number of atypical pairs when the group remains collectively coherent. It is also non-monotone over clique containment, so structural pivot pruning and early rejection of infeasible partial cliques are not directly applicable. We propose two efficient algorithms with different advantageous regimes under this constraint. STRSUB restores structural pivoting by separating structural MCE from bounded semantic subset enumeration. MONOSEMMCE uses a feasible-parent theorem: deleting a deterministic minimum-contribution vertex from any feasible clique preserves feasibility. The induced canonical forest generates each feasible state once without a visited table or materialized size layers. Across three naturally attributed and five controlled vector-attributed graphs, the two specialized methods jointly complete all 24 workloads, whereas the SEMBK baseline reaches the 1,200-second limit on 15. At q_{99} , MONOSEMMCE is $2.92\times-6.24\times$ faster than STRSUB. On two real graphs, pairwise-threshold MCE retains no reference MSC of size at least ten on GitHub-Social and only 0.32% on ogbn-arxiv-25K, demonstrating the model-level fragmentation avoided by the average constraint.

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The source code, data, and/or other artifacts have been made available at <https://github.com/CKTtari/maximal-semantic-clique-enumeration>.

1 INTRODUCTION

Graphs with vector-valued attributes are increasingly common in data management: social networks associate users with profile features, knowledge graphs attach text embeddings to entities, and citation graphs represent papers by content vectors [18, 32, 37]. Existing models combine dense structure with attributes through correlated patterns and similarity-constrained bipartite subgraphs [34, 41]. These tasks expose a basic modeling gap: adjacency establishes interaction, whereas vector similarity establishes semantic compatibility, and neither signal alone identifies groups satisfying both requirements.

The maximal clique is the strongest structural model of cohesion: every pair of vertices is connected. In a vector-attributed graph, this structural contract provides a natural candidate group, and the attributes then determine whether its members are collectively compatible. Partitioning and query-oriented attributed-community methods answer different tasks because they need not preserve pairwise adjacency or enumerate every qualifying group [15, 20, 33]. Our objective is therefore to retain the clique contract while adding a group-level semantic criterion to each reported pattern.

The MSC Problem. We study the problem of enumerating *maximal semantic cliques* (MSCs) in a vector-attributed graph. Each vertex carries a real-valued attribute vector; pairwise semantic similarity is measured by cosine similarity. Given a threshold τ , a clique is a τ -semantic clique if its average pairwise similarity is at least τ . An MSC is an inclusion-wise maximal τ -semantic clique: no strict superset is also a τ -semantic clique. The task is to enumerate *all* MSCs of the input graph.

We use average pairwise similarity to express compatibility at the group level. An every-pair threshold rejects a group whenever one pair falls below the threshold and can fragment a structurally complete group even when the remaining pairs provide strong collective evidence. The average permits a limited number of atypical pairs while expressing the total semantic requirement through a single group-level statistic. This robustness changes the algorithmic problem: adding a vertex may raise or lower the average, and an infeasible clique may have a feasible strict superset. Feasibility is therefore neither downward nor upward closed over clique containment.

For example, a cross-disciplinary co-authorship clique may contain one low-similarity author pair while its remaining pairs provide enough semantic evidence for the complete team to satisfy the average threshold.

Why Classical MCE Does Not Suffice. Classical maximal clique enumeration (MCE) reports only cliques that are maximal under

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adjacency. An MSC, however, may be a proper subset of a structural maximal clique: adding another adjacent vertex preserves the clique but can lower its average similarity below τ . Consequently, filtering the MCE output omits valid MSCs that are never reported as structural maxima. The same distinction invalidates a direct transfer of the Bron–Kerbosch (BK) search [4] and its pivot rule. The pivot suppresses branches whose structural answers can be extended by the pivot, whereas that extension need not remain semantically feasible. Moreover, an infeasible partial clique cannot be discarded because subsequent additions may restore its average. Thus the average constraint changes both the search space and the cost of validating a structural answer: a direct adaptation must explore pivot-suppressed subcliques and postpone threshold-based rejection until semantic feasibility can be established safely.

These observations also imply that no single search organization dominates all thresholds. At permissive thresholds, structural decomposition amortizes semantic work over relatively few host cliques. At selective thresholds, enumerating every structural maximal clique incurs work in regions containing few feasible semantic cliques, while a feasibility-driven search can contract rapidly. Our algorithms are designed around this threshold-dependent change in the search space.

We make the following contributions.

Problem formulation. We formalize exact MSC enumeration under inclusion-wise maximal average similarity, prove the associated size decision problem NP-complete, and characterize why neither structural MCE followed by filtering nor direct BK pivoting is complete for this output family.

Enumeration algorithms. We develop STRSUB, which separates structural MCE from bounded semantic subset enumeration, and MONOSEMCE, which turns the feasible minimum-contribution deletion property into a canonical reverse-search forest. They target permissive and selective thresholds, respectively, while returning the same complete MSC output set.

Large-scale evaluation. We evaluate exactness, runtime, memory, scalability, search-space reduction, and output behavior on three naturally attributed and five controlled vector-attributed graphs under a fixed similarity-quantile protocol. The experiments identify the operating regions of both searches, isolate their mechanisms, and compare the MSC definition with pairwise-threshold and relaxed structural pattern families.

2 PRELIMINARIES

2.1 Notations and Problem Definition

We study a *vector-attributed graph* $\mathcal{G} = (V, E, \mathbf{X})$, where V is the vertex set, $E \subseteq \binom{V}{2}$ is the edge set, and $\mathbf{X} = \{\mathbf{x}_v \in \mathbb{R}^D \mid v \in V\}$ assigns a D -dimensional attribute vector to each vertex.

DEFINITION 1 (SIMILARITY FUNCTION). We instantiate semantic similarity by cosine similarity. After ℓ_2 normalization, $\text{sim}(u, v) = \mathbf{x}_u^\top \mathbf{x}_v$ and $\text{sim}(u, v) \in [-1, 1]$.

Throughout this paper we work with normalized vectors and precompute $\text{sim}(u, v)$ for all $(u, v) \in E$. The threshold $\tau \in [-1, 1]$ is drawn from the same range.

DEFINITION 2 (CLIQUE). A set $C \subseteq V$ is a clique in $\mathcal{G}$ if every pair of distinct vertices in C is connected by an edge, i.e., $\{u, v\} \in E$ for all $u \neq v \in C$.

Classical MCE seeks cliques that cannot be extended while preserving complete connectivity. In a vector-attributed graph, the attributes add a second, group-level question: how similar are the members represented by a structural clique? We express this semantic condition through two companion quantities.

DEFINITION 3 (PAIRWISE SIMILARITY SUM AND AVERAGE PAIRWISE SIMILARITY). For a clique C of size $k = |C| \geq 2$, the pairwise similarity sum is

$$\text{simSum}(C) = \sum_{\{u, v\} \subseteq C} \text{sim}(u, v),$$

and the average pairwise similarity is

$$\text{avgSim}(C) = \frac{\text{simSum}(C)}{\binom{k}{2}}.$$

DEFINITION 4 (SEMANTIC CLIQUE). Given a threshold $\tau \in [-1, 1]$, a clique C with $|C| \geq 2$ is a τ -semantic clique if $\text{avgSim}(C) \geq \tau$.

For algorithmic purposes, we equivalently track the *semantic surplus*

$$W(C) = \text{simSum}(C) - \tau \binom{k}{2},$$

which is non-negative if and only if $\text{avgSim}(C) \geq \tau$. Working with simSum and W avoids division at every recursive step; avgSim is used only when a human-readable quantity is required.

DEFINITION 5 (MSC). Given $\mathcal{G} = (V, E, \mathbf{X})$ and threshold $\tau \in [-1, 1]$, a τ -semantic clique $C \subseteq V$ is a maximal semantic clique (MSC) if there is no τ -semantic clique $C' \supsetneq C$.

Inclusion-wise maximality requires considering all strict supersets; the absence of a feasible single-vertex extension alone is insufficient because average feasibility is non-monotone. For example, let $\tau = 0.6$ and let four vertices form a clique with $\text{sim}(a, b) = 0.6$, $\text{sim}(a, u) = \text{sim}(b, u) = \text{sim}(a, v) = \text{sim}(b, v) = 0.5$, and $\text{sim}(u, v) = 1$. The feasible clique $\{a, b\}$ has no feasible single-vertex extension, since either three-vertex average is $1.6/3 < 0.6$, but the strict superset $\{a, b, u, v\}$ is feasible with average $3.6/6 = 0.6$. Consequently, algorithms may use the absence of a feasible single extension only to generate candidates; a global containment test is still required for MSC output.

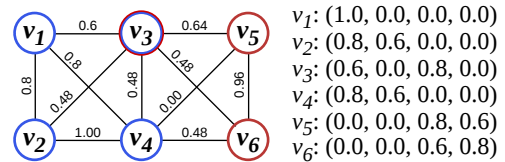


Figure 1: Running example with vector attributes and edge similarities.

EXAMPLE 1. Figure 1 shows a vector-attributed graph with six vertices. Edge labels give the precomputed pairwise similarities. With threshold $\tau = 0.6$, consider the two structural cliques

$$M_1 = \{v_1, v_2, v_3, v_4\}, \quad M_2 = \{v_3, v_4, v_5, v_6\}.$$

Let $C_1 = \{v_1, v_2, v_3, v_4\}$, $C_2 = \{v_3, v_5, v_6\}$, and $C_3 = \{v_3, v_4, v_5, v_6\}$. Their total and average similarities are

$$\begin{aligned} \text{simSum}(C_1) &= 4.16, & \text{avgSim}(C_1) &\approx 0.693 \geq \tau, \\ \text{simSum}(C_2) &= 2.08, & \text{avgSim}(C_2) &\approx 0.693 \geq \tau, \\ \text{simSum}(C_3) &= 2.56, & \text{avgSim}(C_3) &\approx 0.427 < \tau. \end{aligned}$$

Thus C_1 and C_2 are τ -semantic cliques, whereas the structurally valid clique C_3 is not.

Moreover, neither C_1 nor C_2 has a strict superset that is a τ -semantic clique, so both are MSCs.

Problem Statement.

PROBLEM 1 (MSC ENUMERATION). Given a vector-attributed graph $\mathcal{G} = (V, E, X)$ and a threshold $\tau \in [-1, 1]$, enumerate all MSCs of $\mathcal{G}$.

2.2 Challenges

Hardness. MSC enumeration contains classical MCE as a special case. When singleton outputs are excluded and τ does not exceed the minimum similarity on any edge, every structural maximal clique of size at least two satisfies the semantic constraint, so the $\Omega(3^{n/3})$ output lower bound carries over. The associated decision problem is NP-complete.

THEOREM 2.1. Given $\mathcal{G} = (V, E, X)$, threshold $\tau \in [-1, 1]$, and integer k , deciding whether an MSC of size $\geq k$ exists is NP-complete.

PROOF. *NP membership.* The decision instance is positive if and only if it contains a τ -semantic clique C of size at least k . The forward implication is immediate. For the reverse implication, the finite family of feasible clique supersets of C contains an inclusion-wise maximal member, whose size is at least $|C| \geq k$. A certificate therefore need only provide such a feasible clique C : verifying all edges and $\text{avgSim}(C) \geq \tau$ takes $O(|C|^2)$ time. Accordingly, NP verification requires only this feasibility certificate. If maximality were tested separately, examining only single-vertex extensions would be insufficient under the non-monotone constraint.

NP-hardness. We reduce from the CLIQUE decision problem [22], which asks whether $G_0 = (V, E)$ contains a clique of size $\geq k$. Assign $\mathbf{x}_v = \mathbf{e}_1$ to every vertex, so $\text{sim}(u, v) = 1$ for every edge, and set $\tau = 1$. Under this assignment $\text{avgSim}(C) = 1 \geq \tau$ for every clique C , so every structural maximal clique is an MSC and vice versa. Any clique of size $\geq k$ in G_0 is contained in some structural maximal clique of size $\geq k$, which is also an MSC of size $\geq k$. Conversely, any MSC of size $\geq k$ is a clique of size $\geq k$ in G_0 . Hence a size- k MSC exists if and only if G_0 contains a clique of size $\geq k$, completing the reduction. $\square$

The output sets of MSC enumeration and MCE are incomparable: a structural maximal clique with $\text{avgSim}(M) < \tau$ may contain semantically maximal subsets that MCE never generates, while valid MSCs need not be structurally maximal. Therefore, enumerating structural maximal cliques and retaining only those that pass the

average constraint is incomplete because an MSC need not be structurally maximal. The output size, and hence the computational cost, follows the threshold-dependent behavior characterized below.

Output Distribution and Practical Scope. The number and sizes of MSCs vary substantially with τ . At low thresholds, the output approaches classical MCE. As τ increases, infeasible structural cliques can fragment into many incomparable feasible sub-cliques, inflating both the output and the unavoidable enumeration cost. At selective thresholds, most cliques become infeasible and the output contracts. Because these transitions depend on each dataset’s similarity distribution, Section 6 uses a fixed grid of edge-similarity quantiles rather than runtime-selected absolute thresholds.

Non-monotonicity and Framework Incompatibility. The core difficulty is that avgSim is non-monotone over the subset lattice. In classical MCE, structural feasibility is monotone and maximality is enforced implicitly by the $P = \emptyset$ invariant at every leaf, so the set of feasible extensions can be characterized solely by the current state of C . Under the average constraint, no such invariant exists: a partial clique with $\text{avgSim}(C) < \tau$ may still grow into a valid MSC, while one with $\text{avgSim}(C) \geq \tau$ may admit no feasible extension. Furthermore, whether a candidate vertex can participate in a valid extension depends on its accumulated similarity to the current partial clique, which evolves throughout the recursion and cannot be captured by structural conditions alone. As a result, the pruning mechanisms based on structural monotonicity lose their foundation, and the pivot strategy no longer applies. A direct BK adaptation must therefore retain structurally valid intermediate states even when they are currently infeasible. In the baseline developed next, τ validates terminal semantic candidates but does not reduce the structural recursion itself.

3 FRAMEWORK

Framework Overview. SEMBK is a direct BK baseline that retains degeneracy-ordered (P, X) recursion while omitting the structurally justified pivot. At each leaf, a secondary CHECK pass over X enforces inclusion-wise semantic maximality.

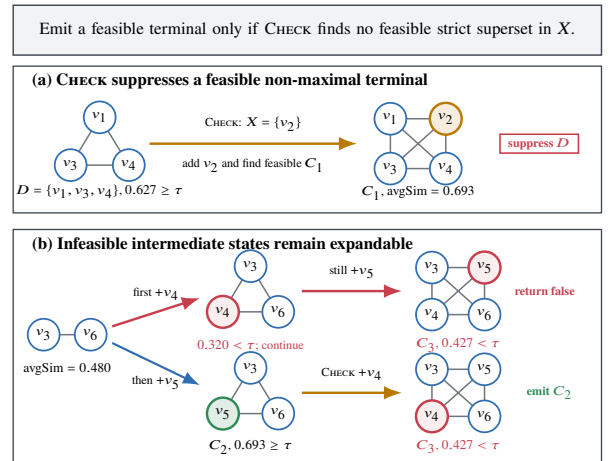


Figure 2: Selected SEMBK branches on the running example.

Algorithm 1: Semantic BK with Check

```
1 Input:  $G = (V, E)$ , semantic vectors  $\{x_v\}$ , threshold  $\tau$ 
2 Output: All MSCs  $\mathcal{R}$ 
3 Precompute  $\text{sim}(u, v)$  and degeneracy ordering  $\sigma$ ;
4 Each worker initializes a private result set  $\mathcal{R}_t \leftarrow \emptyset$ ;
5 for each  $v \in \sigma$  in parallel do
6    $P \leftarrow \{u \in N(v) \mid \sigma(u) > \sigma(v)\}$ ;
7    $X \leftarrow \{u \in N(v) \mid \sigma(u) < \sigma(v)\}$ ;
8   Search( $C = \{v\}, s_C = 0, P, X, \text{mode} = \text{ENUM}, 0$ )
9 return  $\bigcup_t \mathcal{R}_t$ ;

10 Function Search( $C, s_C, P, X, \text{mode}, k_{\min}$ ):
11    $k \leftarrow |C|$ ,  $\text{hasFeasibleDesc} \leftarrow \text{false}$ ;
12    $Q \leftarrow P$ ;
13   for each  $v \in Q$  do
14      $s_D \leftarrow s_C + \sum_{u \in C} \text{sim}(u, v)$ ;
15      $P \leftarrow P \setminus \{v\}, P' \leftarrow P \cap N(v), X' \leftarrow X \cap N(v)$ ;
16      $\text{res} \leftarrow \text{Search}(C \cup \{v\}, s_D, P', X', \text{mode}, k_{\min})$ ;
17     if  $\text{res}$  then
18        $\text{hasFeasibleDesc} \leftarrow \text{true}$ ;
19       if  $\text{mode} = \text{CHECK}$  then return true;
20    $X \leftarrow X \cup \{v\}$ ;
21 if  $\neg \text{hasFeasibleDesc}$  then
22   if  $k \geq 2 \wedge s_C < \tau \binom{k}{2}$  then return false;
23   if  $\text{mode} = \text{CHECK}$  then return  $k > k_{\min}$ ;
24   if  $X \neq \emptyset$  then
25     if Search( $C, s_C, X, \emptyset, \text{CHECK}, k$ ) then
26       return true;
27   if  $k \geq 2$  then  $\mathcal{R}_t \leftarrow \mathcal{R}_t \cup \{C\}$ ;
28   return true;
29 return  $\text{hasFeasibleDesc}$ ;
```

Figure 2 exposes the graph states behind both terminal decisions. In Figure 2(a), the feasible terminal $D = \{v_1, v_3, v_4\}$ has $X = \{v_2\}$; CHECK reconstructs the feasible strict superset C_1 and suppresses D . Figure 2(b) follows the v_6 -owned branch from $C = \{v_6, v_3\}$. The first insertion, v_4 , remains infeasible even after adding v_5 . After backtracking, inserting v_5 yields C_2 with v_4 in its exclusion set. CHECK reaches infeasible C_3 , returns false, and the outer search emits C_2 .

Degeneracy-Root Ownership. Degeneracy ordering assigns each clique to its earliest vertex under σ : later neighbors form P , and earlier neighbors form X . This unique seed owns every clique, including each MSC, while bounding the initial candidate set by graph degeneracy.

Terminal Feasibility and Maximality Check. The Boolean return value records whether a subtree contains a feasible terminal. Only when no child reports one does C become a candidate; an infeasible candidate is discarded. A feasible candidate can still have a strict extension through X , so CHECK searches that exclusion set and suppresses C upon finding one. The parameter $k_{\min} = |C|$ enforces strict containment because the check recursion may also revisit sets no larger than C .

Why pivoting is not used. In standard BK, a pivot $p \in P \cup X$ guarantees that each structural maximal clique in the current branch contains either p or a vertex in $P \setminus N(p)$. Hence vertices in $P \cap N(p)$ need not become expansion roots. This argument depends on structural feasibility being preserved when an adjacent pivot is added.

The average constraint invalidates that implication. Consider a complete graph on $\{s, p, u, w\}$ at the BK state $C = \{s\}, P = \{p, u, w\}$, and $X = \emptyset$. Let the three similarities within $\{s, u, w\}$ equal 0.9, the three similarities incident to p equal 0, and $\tau = 0.6$. Then $\{s, u, w\}$ is an MSC with average 0.9, whereas its only strict clique superset has average 0.45. Choosing p as the pivot leaves only p in $P \setminus N(p)$, so every explored child contains p and the valid MSC $\{s, u, w\}$ is lost. Pivot-free branching is therefore required for the direct semantic BK baseline.

Complexity.

THEOREM 3.1 (TIME AND SPACE COMPLEXITY). *Let d_G be the degeneracy of G , let $\mathcal{L}$ be the feasible terminal states reached by the outer search, and let X_C be the candidate set examined by CHECK for $C \in \mathcal{L}$. The running time is $O^*(n2^{d_G} + \sum_{C \in \mathcal{L}} 2^{|X_C|})$, where O^* suppresses polynomial set-intersection costs. A graph-independent upper bound is $O^*(3^n)$. With T workers, the implementation uses $O(m + nD + Tn^2)$ worst-case space.*

PROOF. A degeneracy seed has at most d_G later neighbors. Without pivoting, its outer recursion may visit every subset of those neighbors, giving $O^*(n2^{d_G})$ states. For a feasible terminal C , CHECK may independently visit every subset of X_C . The same earlier extension can be reconsidered for several terminals; charging a state to a pair of nested vertex sets gives the safe $O^*(3^n)$ bound. Similarities and sparse adjacency require $O(m + nD)$ storage. Each worker owns an $O(n)$ accumulator, while decreasing candidate arrays retained by a worst-case CHECK stack total $O(n^2)$. $\square$

For comparison, degeneracy-oriented pivoted MCE has the time bound [11, 12]

$$O(d_G n 3^{d_G/3})$$

The average constraint introduces more than an $O(|C|^2)$ validation cost: valid MSCs may occur at nonmaximal structural states, so the direct search may inspect all 2^{d_G} subsets below a seed and revisit earlier-neighbor subsets in CHECK. This gap formalizes the additional search complexity relative to structural MCE.

Design Implication. SEMBK is a direct same-definition baseline that uses local search state and no global visited table. Its pivot-free main search cannot prune every infeasible intermediate, while CHECK may revisit earlier-neighbor subspaces. Section 4 addresses these costs with two search organizations tailored to different threshold regimes.

4 PROPOSED ALGORITHMS

This section presents two exact algorithms for complementary threshold regimes. STRSUB decouples structural enumeration from semantic subset processing, retaining Tomita pivoting and using admissible size-layer bounds. MONOSEMMCE uses canonical reverse search, allowing τ to reject infeasible child states before recursion. We evaluate both methods on the same quantile grid: STRSUB

provides broad threshold coverage, while MONOSEMCE targets selective high- τ workloads.

4.1 STRSUB: Structural Decomposition and Subset Enumeration

Structural–Semantic Decoupling and Containment. STRSUB restores the Tomita pivot by separating structural enumeration from semantic subset search. Pivoted BK first produces structural maximal cliques; the average constraint is then evaluated only within each resulting host. Lemma 4.1 establishes that this separation preserves every MSC.

For a host clique M and target size k , let $\bar{e}_k$ be the mean of the $\binom{k}{2}$ largest pairwise similarities in M . For each $v \in M$, let $q_v(k)$ be the sum of its $k-1$ largest incident similarities within M , and let $\bar{q}_k$ be the sum of the k largest $q_v(k)$ values divided by $k(k-1)$. Both quantities upper-bound the average similarity of any size- k subset; Properties 4.1 and 4.2 prove these bounds below.

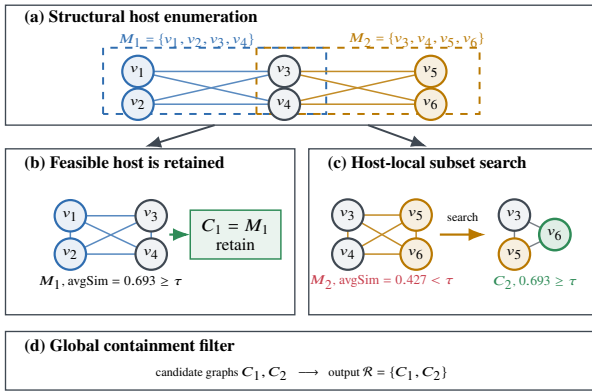


Figure 3: STRSUB execution on the running example.

Figure 3(a) first exposes the two overlapping structural hosts $M_1 = \{v_1, v_2, v_3, v_4\}$ and $M_2 = \{v_3, v_4, v_5, v_6\}$. The semantic phase handles each host independently. Since M_1 satisfies the average constraint, Figure 3(b) retains it directly as C_1 . The infeasible M_2 instead activates the bounded subset search in Figure 3(c), which recovers the feasible triangle $C_2 = \{v_3, v_5, v_6\}$. The global containment filter then removes duplicate candidates and strict subsets before returning $\mathcal{R} = \{C_1, C_2\}$ (Figure 3(d)).

LEMMA 4.1 (STRUCTURAL SUPERSET OF MSCs). *Every MSC is contained in at least one structural maximal clique of G . That is, for every MSC C , there exists a structural maximal clique M such that $C \subseteq M$.*

PROOF. By definition, every MSC is a clique of G . Let C be an arbitrary MSC, and consider the family

$$\mathcal{F} = \{D \subseteq V \mid C \subseteq D \text{ and } D \text{ is a clique in } G\}.$$

Since $C \in \mathcal{F}$, the family is nonempty. Because V is finite, $\mathcal{F}$ is finite, so we may choose a set $M \in \mathcal{F}$ that is maximal under inclusion. By construction, M is a clique containing C , and no vertex $w \in V \setminus M$ extends M to a larger clique; hence M is a structural maximal clique of G , and $C \subseteq M$. $\square$

Algorithm 2: STRSUB: Structural Decomposition and Subset Enumeration

Input: $G = (V, E)$, similarities $\{\text{sim}(u, v)\}_{(u,v) \in E}$, threshold τ
Output: All MSCs $\mathcal{R}$

```

1  $\mathcal{F} \leftarrow \emptyset$ ;
  /* Pipelined structural MCE and semantic processing */
2 Build degeneracy ordering  $\sigma$ ;
3 for each  $v \in \sigma$  in parallel do
4    $P \leftarrow \{u \in N(v) \mid \sigma(u) > \sigma(v)\}$ ;
5    $X \leftarrow \{u \in N(v) \mid \sigma(u) < \sigma(v)\}$ ;
6   Run BK( $\{v\}, P, X$ ) with the following callback at each leaf  $M$ ;
  /* Callback: intra-clique semantic enumeration */
7 for each structural leaf  $M$  produced by a worker do
8   if  $\text{avgSim}(M) \geq \tau$  then  $\mathcal{F} \leftarrow \mathcal{F} \cup \{M\}$ ; continue;
  /* Admissible layer bounds */
9   Sort edges of  $M$  by similarity descending;
10   $k^* \leftarrow \max\{k \in [2, |M| - 1] \mid \bar{e}_k \geq \tau\}$ ;
11  if  $k^*$  does not exist then continue;
12   $lmax \leftarrow \emptyset$ ;  $umask \leftarrow \emptyset$ ;
13  for  $k \leftarrow k^*$  to 2 do
14    Compute the vertex-contribution bound  $\bar{q}_k$  for size  $k$ ;
15    if  $\bar{q}_k < \tau$  then continue;
16    for each size- $k$  subset  $S \subseteq M$  do
17      if  $(S \& umask) = S$  and  $\exists T \in lmax : S \subseteq T$  then
18        continue;
19      if  $\text{avgSim}(S) \geq \tau$  then
20         $lmax \leftarrow lmax \cup \{S\}$ ;
21         $umask \leftarrow umask \mid S$ ;
22         $\mathcal{F} \leftarrow \mathcal{F} \cup \{S\}$ ;
  /* Phase 3: Exact global filter */
23 Sort  $\mathcal{F}$  by  $|\cdot|$  descending; remove exact duplicates;
24 Initialize an empty inverted index  $inv : v \mapsto \emptyset$ ;
25  $\mathcal{R} \leftarrow \emptyset$ ;
26 for each equal-size group  $\mathcal{G} \subseteq \mathcal{F}$  in descending size do
27   for each  $S \in \mathcal{G}$  in parallel do
28      $r \leftarrow \arg \min_{v \in S} |inv[v]|$ ;
29     Keep  $S$  iff no  $T$  referenced by  $inv[r]$  satisfies  $S \subseteq T$ ;
30 Insert every kept  $S \in \mathcal{G}$  into  $\mathcal{R}$  and  $inv$ ;
31 return  $\mathcal{R}$ ;

```

Structural Host Enumeration. Phase 1 uses degeneracy-oriented, pivoted BK [12, 36] and is independent of τ . Degeneracy roots are dynamically scheduled; upon reaching a structural leaf M , the owning worker immediately invokes the semantic callback instead of materializing the complete family of structural maximal cliques.

Bounded Semantic Subset Search. For each host M , Phase 2 visits subset sizes in descending order and uses a host-local similarity matrix for repeated feasibility tests. The local filter stores previously found feasible sets in $lmax$ and uses $umask = \bigcup_{T \in lmax} T$ as a fast prerequisite for an exact containment probe. If $S \subseteq T$ for feasible T , then S cannot be an MSC; the filter therefore removes S without changing the output. The admissible bounds below skip entire size layers before subset generation.

Global Maximality Reconciliation. An MSC may occur in multiple structural maximal cliques, so semantic subset search may emit duplicate or subsumed candidates. Global reconciliation consolidates them into $\mathcal{R}$. Candidates are deduplicated and processed by decreasing size. For each S , the inverted index is probed through its rarest vertex, limiting exact containment tests to retained supersets that share that vertex. Equal-size queries read a frozen index in parallel and are inserted only after the group completes. Because equal-size sets cannot strictly contain one another and discarded supersets are covered transitively by retained ones, this grouped procedure is equivalent to testing all larger candidates.

Admissible Size-Layer Bounds. Phase 2 as described iterates over all subset sizes from $|M| - 1$ down to 2. A key observation allows the starting size to be reduced: the best achievable avgSim for any size- k subset of M is bounded above by the mean of the $\binom{k}{2}$ largest pairwise similarities in M .

PROPERTY 4.1 (SORTED-EDGE UPPER BOUND). *Let $e_1 \geq e_2 \geq \dots \geq e_{\binom{m}{2}}$ be the pairwise similarities of M sorted in descending order. For any size- k subset $S \subseteq M$,*

$$\text{avgSim}(S) \leq \bar{e}_k := \frac{1}{\binom{k}{2}} \sum_{i=1}^{\binom{k}{2}} e_i.$$

PROOF. Any size- k subset $S \subseteq M$ uses exactly $\binom{k}{2}$ pairs, each contributing at most the corresponding value in the sorted sequence $e_1 \geq e_2 \geq \dots$; hence $\text{simSum}(S) \leq \sum_{i=1}^{\binom{k}{2}} e_i$, and dividing both sides by $\binom{k}{2}$ gives the result. $\square$

The sorted-edge bound ignores that its largest edges may be concentrated on too few vertices to coexist in one size- k subset. A complementary bound tracks which vertices support those similarities. For each $v \in M$, let $q_v(k)$ be the sum of the $k - 1$ largest similarities between v and vertices in $M \setminus \{v\}$. Let $q_{(1)}(k) \geq \dots \geq q_{(m)}(k)$ denote these values in descending order.

PROPERTY 4.2 (VERTEX-CONTRIBUTION UPPER BOUND). *For any size- k subset $S \subseteq M$,*

$$\text{avgSim}(S) \leq \bar{q}_k := \frac{1}{k(k-1)} \sum_{i=1}^k q_{(i)}(k).$$

PROOF. For every $v \in S$, the sum of its similarities to the other vertices of S uses $k - 1$ vertices from $M \setminus \{v\}$ and is therefore at most $q_v(k)$. Summing over S counts each internal pair twice, so

$$2\text{simSum}(S) \leq \sum_{v \in S} q_v(k) \leq \sum_{i=1}^k q_{(i)}(k).$$

Division by $k(k-1)$ gives the claimed average-similarity bound. $\square$

The prefix means $\bar{e}_k$ are non-increasing in k . The algorithm identifies

$$k^* = \max\{k \in [2, |M| - 1] \mid \bar{e}_k \geq \tau\},$$

and skips every larger layer. For each remaining k , it also skips the layer when $\bar{q}_k < \tau$. Both bounds are admissible independently, so their joint use cannot remove a feasible subset. Sorting the edge and per-vertex similarity lists costs $O(m^2 \log m)$ per host.

Applicable Regime. STRSUB applies Tomita pivoting during structural MCE, whose cost is $O(d_G n 3^{d_G/3})$ and independent of τ . At low thresholds, large feasible subsets are identified early within each structural host. At higher thresholds, the sorted-edge and vertex-contribution bounds skip infeasible size layers. Structural MCE remains a fixed phase cost, which motivates the canonical search developed next for selective thresholds.

Complexity.

THEOREM 4.1 (TIME COMPLEXITY OF STRSUB). *Let $\mathcal{M}$ denote the family of all structural maximal cliques of G . For each $M \in \mathcal{M}$, let Q_M denote the number of exact local containment probes and let $b_M = \lceil |M|/w \rceil$ be its bitmap length in machine words. STRSUB runs in*

$$O\left(d_G n 3^{d_G/3} + \sum_{M \in \mathcal{M}} \left(2^{|M|} |M|^2 + Q_M b_M\right) + KN \log N + K \sum_{S \in \mathcal{F}} L_{\min}(S)\right)$$

time, where d_G is the degeneracy of G , $N = |\mathcal{F}|$ is the number of candidates before global deduplication, $K = \max_{S \in \mathcal{F}} |S|$, and $L_{\min}(S) = \min_{v \in S} |\text{inv}[v]|$.

PROOF. The structural phase uses degeneracy-oriented Tomita BK [12], with cost $O(d_G n 3^{d_G/3})$. For a host M of size h , subset enumeration visits at most 2^h subsets and evaluates each from the cached similarity matrix in $O(h^2)$ time. Exact local maximality performs Q_M bitmap containment probes of b_M words each; keeping this term explicit also covers instances with many feasible local subsets. The final filter sorts N candidates in $O(KN \log N)$ time. It then scans the shortest inverted list for each S , with at most $L_{\min}(S)$ sorted-set containment checks of $O(K)$ time. $\square$

Within a host, non-monotone feasibility retains the exponential worst case; the admissible bounds and local containment reduce inspected candidates without changing the output.

4.2 MONOSEMMCE: High- τ Canonical Reverse Search

Motivation and Core Insight. Even with the size-layer bounds, STRSUB is bound to Phase 1: it enumerates all structural maximal cliques before any semantic evaluation takes place. In the high- τ regime, the vast majority of structural cliques may contain no feasible semantic result, yet each still contributes Phase 1 cost. More fundamentally, the BK framework offers no mechanism to incorporate τ into branching decisions—the threshold appears only in validation, never in the search logic itself. This structural overhead motivates abandoning the BK skeleton entirely.

The obstacle to integrating τ directly into search is the non-monotonicity of avgSim: an infeasible clique can have a feasible strict superset. A search that retains only feasible states is therefore complete only if every feasible target has a feasible construction path. Theorem 4.2 supplies such a path through the deletion rule defined next.

For a vertex v in a clique C , define its *similarity contribution* as $a_C(v) = \sum_{u \in C \setminus \{v\}} \text{sim}(u, v)$: the sum of its similarities to the other vertices in C . We delete a vertex with minimum similarity contribution, breaking ties by vertex identifier. The theorem below proves

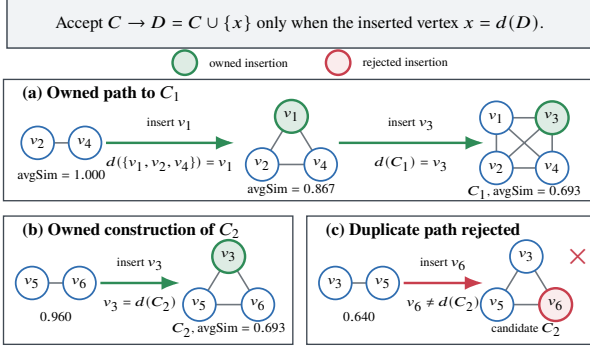


Figure 4: Canonical ownership on the running example.

that the remaining clique is feasible; the deterministic deletion therefore defines the canonical parent used by reverse search.

The implementation traverses canonical trees by DFS from feasible edge roots. It stores no complete size layer; a final containment filter enforces inclusion-wise maximality.

Feasible Parents and the Canonical Forest.

THEOREM 4.2 (FEASIBLE MINIMUM-CONTRIBUTION DELETION). *Let C be a τ -semantic clique of size $k \geq 3$, with the similarity contribution $a_C(v)$ defined above. If $d(C)$ minimizes the pair $(a_C(v), v)$ lexicographically, then $C \setminus \{d(C)\}$ is a τ -semantic clique. Repeated application of this rule therefore reaches a feasible edge, and reversing these deletions gives a non-increasing-average insertion path.*

PROOF. Each pair is counted once at each endpoint. The minimum contribution therefore satisfies $a_C(d(C)) \leq 2\text{simSum}(C)/k$. Deleting $d(C)$ leaves similarity sum at least $\text{simSum}(C)(k-2)/k$, and hence

$$\text{avgSim}(C \setminus \{d(C)\}) \geq \frac{\text{simSum}(C)(k-2)/k}{\binom{k-1}{2}} = \text{avgSim}(C) \geq \tau.$$

Deterministic tie-breaking makes $d(C)$ unique. Applying the same argument until two vertices remain preserves feasibility at every deletion; reversing the sequence proves the insertion-path claim. $\square$

Define $p(C) = C \setminus \{d(C)\}$ for every feasible clique C of size at least three. The theorem makes $p(C)$ a feasible and unique parent, so all feasible cliques form a forest rooted at feasible edges. This property does not make arbitrary insertions or deletions monotone; it identifies one complete feasible path for every target. MONO-SEMMCE traverses precisely these owned parent-child edges and rejects alternative constructions before recursion.

Figure 4 applies this rule to the running example. The upper sequence constructs C_1 through canonical insertions. The lower panels contrast the owned construction of C_2 with an alternative insertion that forms the same clique but is rejected before recursion because the inserted vertex is not $d(C_2)$.

Feasible Edge Roots. Every feasible clique of size at least two reaches a feasible edge under the deterministic deletion rule, so Algorithm 3 starts one tree for each edge with similarity at least τ . A state stores the clique C , its pairwise sum $s_C = \text{simSum}(C)$,

Algorithm 3: MONOSEMMCE: Canonical Reverse Search

```

1 Input:  $G = (V, E)$ , semantic vectors  $\{x_v\}$ , threshold  $\tau$ 
2 Output: All MSCs  $\mathcal{R}$ 
3 Precompute  $\text{sim}(u, v)$  for all  $(u, v) \in E$ ;
4  $\mathcal{F} \leftarrow \emptyset$ ;
5 for each  $(u, v) \in E$  with  $u < v$  and  $\text{sim}(u, v) \geq \tau$  in parallel do
6    $C \leftarrow \{u, v\}$ ;  $s_C \leftarrow \text{sim}(u, v)$ ;
7    $P \leftarrow N(u) \cap N(v)$ ;
8    $\text{acc}[w] \leftarrow \text{sim}(w, u) + \text{sim}(w, v)$  for  $w \in P$ ;
9    $a_C(u) = a_C(v) \leftarrow \text{sim}(u, v)$ ;
10  Search( $C, s_C, P, \text{acc}, a_C$ );
11 Apply Algorithm 2's grouped strict-containment filter to  $\mathcal{F}$ ,
    obtaining  $\mathcal{R}$ ;
12 return  $\mathcal{R}$ ;

13 Function Search( $C, s_C, P, \text{acc}, a_C$ ):
14    $k \leftarrow |C|$ ;  $w_C \leftarrow s_C - \tau \binom{k}{2}$ ;
15    $\text{hasValid} \leftarrow \text{false}$ ;
16   for each  $v \in P$  do
17      $\delta \leftarrow \text{acc}[v]$ ;
18     if  $w_C + \delta - \tau k < 0$  then continue;
19      $\text{hasValid} \leftarrow \text{true}$ ;
20      $D \leftarrow C \cup \{v\}$ ;
21     Update  $a_D$  from  $a_C$  and  $\text{sim}(v, u)$  for  $u \in C$ ;
22     if  $v \neq \arg \min_{x \in D} (a_D(x), x)$  then continue;
23      $P' \leftarrow P \cap N(v)$ ;
24      $\text{acc}'[w] \leftarrow \text{acc}[w] + \text{sim}(w, v)$  for  $w \in P'$ ;
25     Search( $D, s_C + \delta, P', \text{acc}', a_D$ );
26   if not  $\text{hasValid}$  then  $\mathcal{F} \leftarrow \mathcal{F} \cup \{C\}$ ;

```

the common-neighbor set P , candidate accumulators acc , and the member contributions a_C . No BK exclusion set is required: uniqueness follows from the canonical-parent test rather than from an insertion order.

Surplus and Canonical-Child Tests. For $k = |C|$ and $\delta = \text{acc}[v]$, the extension satisfies

$$W(C \cup \{v\}) = W(C) + \delta - \tau k.$$

If this value is negative, the child itself is infeasible and cannot lie on the canonical path of any feasible answer, so it is safely skipped. For a feasible child $D = C \cup \{v\}$, the algorithm updates every member's similarity contribution and accepts D only if $v = d(D)$. Equivalently, the canonical parent of D must be exactly C . This test removes alternative insertion paths before recursion rather than after state materialization. The flag hasValid is set before the ownership test because it records the existence of any feasible one-vertex extension, not the existence of an owned child. A state with such an extension cannot itself be inclusion-wise maximal; the extension is generated from its unique canonical parent.

The canonical test removes duplicate insertion paths before recursion. Accumulator values are restored during backtracking, so the result does not depend on the order of floating-point additions.

Inclusion-Wise Maximality. A state with no feasible single-vertex extension is added to $\mathcal{F}$, but is not yet declared an MSC. If such a candidate C is not inclusion-wise maximal, the finite family

of feasible clique supersets of C contains an inclusion-wise maximal member M . The canonical search enumerates M exactly once, and M has no feasible single extension, so it also belongs to $\mathcal{F}$. The final grouped inverted-index containment filter consequently removes C . Conversely, an MSC is not contained in any other feasible candidate and is retained.

THEOREM 4.3 (ENUMERATION CORRECTNESS). *Algorithm 3 generates every τ -semantic clique of size at least two exactly once and returns exactly all MSCs.*

PROOF. Existence follows by reversing the repeated deterministic deletions in Theorem 4.2; every prefix is feasible and therefore passes the surplus test. Uniqueness follows because every non-root feasible clique has the single parent $C \setminus \{d(C)\}$, and only that parent accepts it as a child. The preceding containment argument proves that the final filter retains precisely the inclusion-wise maximal feasible cliques. $\square$

Complexity.

THEOREM 4.4 (TIME AND SPACE COMPLEXITY OF MONOSEMMCE). *Let F_τ be the number of feasible cliques of size at least two, K the maximum clique size, Δ the maximum degree, T the number of workers, and $N = |\mathcal{F}|$ the number of terminal candidates. A conservative bound for canonical search is*

$$O\left(F_\tau \Delta (\Delta + K \log \Delta) + KN \log N + K \sum_{S \in \mathcal{F}} L_{\min}(S)\right),$$

with $O(m + T(n + K\Delta) + NK)$ words of space in addition to vector storage.

PROOF. Canonical ownership bounds accepted states by F_τ . A state tests at most Δ candidates. For each candidate, common-neighbor intersection and accumulator maintenance cost $O(\Delta)$ conservatively, while updating and comparing at most K member contributions uses cached similarities with $O(\log \Delta)$ lookup. This is bounded by $O(\Delta(\Delta + K \log \Delta))$ per accepted state. The final filter has the same $O(KN \log N + K \sum_S L_{\min}(S))$ cost as in STRSUB.

Sparse similarity adjacency and neighbor bitmaps use $O(m)$ words. Each worker owns an $O(n)$ generation-tagged accumulator and $O(K\Delta)$ DFS scratch space. Terminal candidates occupy $O(NK)$ words; no feasible-size layer or visited-state table is materialized. $\square$

Canonical search is governed by the feasible-state family F_τ rather than structural maximal cliques. This family can exceed the structural output at intermediate thresholds but contracts sharply as τ becomes selective; STRSUB instead continues to pay the structural MCE cost.

5 IMPLEMENTATION

All methods use a common C++17 loader, timing policy, and edge-similarity cache, while their adjacency representation follows the access pattern of each search. STRUCTBK and STRSUB repeatedly evaluate pivot neighborhoods and therefore store each row as a sparse blocked bitmap: only nonempty 64-bit blocks are kept, and intersections merge block identifiers before applying word-level

AND. MONOSEMMCE retains the same representation for common-neighbor construction at feasible edge roots. In contrast, the pivot-free SEMBK scans its evolving candidate and exclusion lists against sorted sparse adjacency rows; on these short, irregular probes, constructing and searching block indices adds work. The three semantic methods place vectors in a degree-descending breadth-first layout for similarity-cache construction while preserving graph vertex IDs.

Similarity Representation. Input vectors are normalized once into 64-byte-aligned float32 arrays, and Advanced Vector Extensions 2 (AVX2) dot products populate the sparse edge-similarity cache shared by all semantic methods. Similarity sums, semantic surplus, and threshold products use double precision, and all algorithms apply the same inclusive comparison $\text{simSum}(C) \geq \tau \binom{|C|}{2}$ and no method-specific tolerance. Within a structural host M , STRSUB materializes an $|M| \times |M|$ float32 matrix so that repeated subset tests avoid graph-level similarity lookups. End-to-end timing includes this preprocessing, search, and result consolidation; optional text serialization is disabled, while phase counters separately report search work.

Incremental Search State. MONOSEMMCE maintains $\text{acc}[w] = \sum_{u \in C} \text{sim}(u, w)$ because every child-feasibility test reads this marginal contribution. Extending C by v updates only surviving candidates as $\text{acc}'[w] = \text{acc}[w] + \text{sim}(w, v)$, reducing an $O(|P'| |C|)$ recomputation to $O(|P'|)$. Each worker owns a generation-tagged array and reuses depth-indexed buffers; logical resets avoid clearing $O(n)$ entries, and backtracking restores modified values. SEMBK instead computes the marginal sum directly from C ; its terminal-only semantic test provides no pruning rule that would consume a per-candidate accumulator. STRSUB evaluates semantic subsets against the host-local float32 matrix and reuses its size-layer buffers.

Parallel Ownership. OpenMP scheduling follows disjoint units already established by the algorithms. STRUCTBK, SEMBK, and STRSUB dynamically assign degeneracy roots to workers. For a large fixed-size subset layer, STRSUB further partitions combinations by their smallest selected vertex; workers read the frozen larger-size local maxima and merge their same-size results after the layer. MONOSEMMCE dynamically assigns feasible edge roots, each of which owns one canonical tree. Mutable recursion buffers, counters, and candidate output remain thread-local during search.

Result Consolidation. Both proposed methods use the grouped containment filter from Algorithm 2. Candidates are sorted once by decreasing size and deduplicated lexicographically. Queries within one equal-size group read a frozen inverted index and run in parallel; retained candidates are inserted only after the group completes. Consequently, synchronization occurs at result merging and between size groups rather than on recursive search states.

6 EXPERIMENTS

Seven research questions cover end-to-end efficiency and output agreement (RQ1), threshold and graph-size scalability (RQ2–RQ3), parallelism and mechanisms (RQ4–RQ5), the MSC output space (RQ6), and definition and pattern-family comparisons (RQ7).

Dataset	$ V $	$ E $	D	Attributes
FB15K-237	14,541	248,611	384	natural
GitHub-Social	37,700	289,003	4,005	natural
ogbn-arxiv	169,343	1,157,799	128	natural
email-Enron	36,692	183,831	256	generated
sc-nasasrb	54,870	1,311,227	256	generated
sc-pkustk11	87,804	2,565,054	256	generated
sc-pwtk	217,891	5,653,221	256	generated
sc-lldoor	909,537	20,770,807	256	generated

Table 1: Public graphs and vector attributes used in the evaluation.

6.1 Experimental Setup

Methods and comparison scope. STRUCTBK implements BK with Tomita pivoting and degeneracy-ordered roots [4, 11, 12, 36]; it supplies a threshold-independent structural-time reference. SEMBK is the direct baseline: it uses the same MSC definition, threshold, similarity values, and strict-containment filter as STRSUB and MONOSEMMCE. Consequently, runtime speedups are claimed only among these three exact MSC methods. PAIRSIM-MCE keeps edges whose endpoint similarity reaches τ and then runs structural MCE; it represents the every-pair similarity contract used by related constrained enumeration models [41]. FastQC enumerates maximal quasi-cliques under a degree-density constraint [42], while FaPlex enumerates maximal 2-plexes that may omit one incident edge per vertex [44]. These methods define different pattern families and are used only for output-family comparisons. **Datasets.** The evaluation contains three graphs with natural attributes and five public graphs with reproducibly generated attributes. FB15K-237 descriptions use all-MiniLM-L6-v2 embeddings; GitHub-Social uses its official 4,005-dimensional developer features; and ogbn-arxiv retains its official 128-dimensional node features [18, 32, 37]. For email-Enron and the four SuiteSparse graphs [9, 24], the PCG64 pseudorandom generator draws 256 coordinates independently from $U[-1, 1)$ under fixed dataset seeds, followed by ℓ_2 normalization. Every graph topology is extracted directly from its public source dataset. Generated vectors provide controlled inputs for runtime, scale, and mechanism experiments; the semantic-output analyses in RQ7 use only the two graphs with official natural attributes.

Protocol. Experiments run on an AMD Ryzen 9 7945HX workstation with 16 physical cores, 32 hardware threads, and 32 GiB memory. We compile with GCC 13.2 and OpenMP. Each process is limited to 16 GiB and 32 OpenMP threads. The uniform timeout is 1,200 seconds. Completed runtime configurations are repeated three times and reported by the median; time-limit exceeded (TLE) denotes one complete 1,200-second limit window. Runtime binaries contain no statistics counters, and the reported in-memory end-to-end time excludes optional text serialization.

Absolute cosine values are not comparable across vector sources. We therefore derive thresholds from the edge-similarity distribution of each dataset. The primary grid fixes q_{20} , q_{80} , and q_{99} before execution; sensitivity experiments use q_5 , q_{20} , q_{50} , q_{70} , q_{80} , q_{90} , q_{95} , q_{97} , and q_{99} .

6.2 RQ1: Exactness and End-to-End Efficiency

This experiment answers RQ1 by comparing complete MSC output sets and total runtime across the 24 fixed dataset-threshold workloads. Exhaustive oracles first test 225,000 vertex-bound instances and 40,001 canonical-parent instances, including negative similarities, equality at τ , and tied minimum contributions. On real graphs, jointly completed methods agree on normalized clique sets, counts, and size histograms. In particular, all three exact methods return 263,721, 71,118, and 2,636 MSCs on GitHub-Social at q_{20} , q_{80} , and q_{99} , respectively; STRSUB and MONOSEMMCE also agree on every jointly completed row of Table 2. Across the eight graphs, the structural reference enumerates between 22,612 and 1,041,789 structural maximal cliques; these counts quantify the structural enumeration workload underlying the runtime comparisons.

Table 2 shows that STRSUB completes 22 of 24 workloads and every q_{20} row, while MONOSEMMCE resolves the two intermediate rows where host-clique subset enumeration reaches TLE. At q_{99} , canonical search is faster on all eight graphs, with speedups from 2.92 $\times$ on sc-lldoor to 6.24 $\times$ on email-Enron. SEMBK completes all nine rows on GitHub-Social, ogbn-arxiv, and email-Enron but reaches TLE on all three FB15K-237 rows; its pivot-free search is slower than the faster of STRSUB and MONOSEMMCE on every completed row. The STRUCTBK column is reported only to separate structural enumeration cost from semantic processing. Its 0.090–1.679 second range is much smaller than the hardest semantic runs, showing that the dominant work comes from enforcing average feasibility and inclusion maximality within the structural search space. The jointly completed rows therefore establish both exact output agreement and the intended division of labor: structural decomposition is effective when entire host cliques or large size layers can be resolved together, whereas canonical traversal is effective after semantic selectivity has contracted the feasible-state forest.

6.3 RQ2: Threshold Sensitivity

This experiment varies semantic selectivity while holding each graph fixed, so a reversal in runtime ordering can be attributed to the threshold-induced search space. Figure 5 uses the full fixed quantile grid. On ogbn-arxiv, STRSUB remains between 0.843 and 1.233 seconds across the sweep, whereas MONOSEMMCE decreases from 3.357 seconds at q_{20} to 0.435 at q_{95} and 0.197 at q_{99} ; the methods cross near q_{95} . On sc-lldoor, STRSUB completes the low-threshold cases, but the intermediate fragmentation region produces 66.6 million MSCs at q_{70} and exhausts its host-clique subset search. As q increases further, infeasible insertions truncate the canonical forest, and MONOSEMMCE falls from 91.945 seconds at q_{70} to 1.658 seconds at q_{99} . These trajectories match the mechanisms developed in Sections 4.1 and 4.2: STRSUB amortizes decisions over structural hosts, while MONOSEMMCE visits only feasible, canonically owned states. TLE points are shown at the fixed limit and are not interpolated.

6.4 RQ3: Scalability with Graph Size

Figure 6 reports absolute end-to-end runtime on shared logarithmic axes as the graph grows through nested induced ogbn-arxiv graphs whose vertices follow one fixed degree order. The construction changes the number of vertices and outputs while preserving the

Dataset	STRUCTBK structure	q	MSCs	SEMBK	STRSUB	MONOSEMMCE
FB15K-237	0.103	20	206,470	TLE	0.221	TLE
		80	76,106	TLE	9.132	TLE
		99	7,476	TLE	45.234	10.091
GitHub-Social	0.110	20	263,721	34.777	0.327	29.315
		80	71,118	27.548	0.314	20.551
		99	2,636	21.495	0.391	0.118
ogbn-arxiv	0.309	20	925,239	1.590	1.233	3.357
		80	226,936	1.165	0.872	1.610
		99	10,261	1.249	0.843	0.197
email-Enron	0.090	20	223,703	9.196	0.259	16.739
		80	60,352	8.312	1.266	0.083
		99	1,819	4.084	0.206	0.033
sc-nasasrb	0.090	20	28,154	TLE	0.301	TLE
		80	834,914	TLE	102.820	0.759
		99	12,742	TLE	0.415	0.079
sc-pkustk11	0.143	20	22,612	TLE	0.645	TLE
		80	2,214,411	TLE	TLE	2.219
		99	24,891	TLE	0.611	0.167
sc-pwtk	0.286	20	58,523	TLE	1.093	TLE
		80	4,483,887	TLE	TLE	3.977
		99	54,907	TLE	1.364	0.421
sc-ldoor	1.679	20	274,585	TLE	4.870	TLE
		80	13,473,317	TLE	563.996	14.944
		99	202,030	TLE	4.845	1.658

Table 2: Median in-memory end-to-end runtime across 24 dataset-threshold workloads.

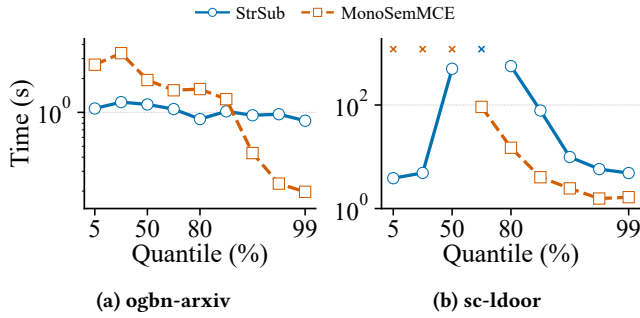


Figure 5: Threshold sweeps reveal complementary runtime regimes.

graph family and all six threshold quantiles. Both exact methods remain executable across the four sizes: at q_{20} , STRSUB increases from 0.516 to 1.216 seconds and MONOSEMMCE from 2.130 to 3.320 seconds; at q_{99} , the corresponding increases are 0.340 to 0.707 and 0.057 to 0.200 seconds. Thus the experiment directly measures how each search organization responds to graph growth, rather than treating one method as a universal reference. Figure 7 complements these absolute curves with the runtime ratio across all size-threshold combinations. Its sign boundary remains at q_{95} : STRSUB is faster through q_{90} , while MONOSEMMCE becomes faster at q_{95} and q_{99} . The latter’s q_{99} advantage is $6.0\times$, $5.6\times$, $4.4\times$, and $3.5\times$ at 25K, 50K, 100K, and 169K vertices, respectively, so the selective-threshold benefit persists as the graph grows by $6.8\times$.

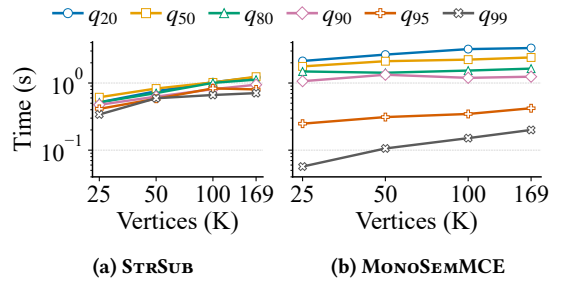


Figure 6: Absolute runtime growth for both methods on logarithmic axes.

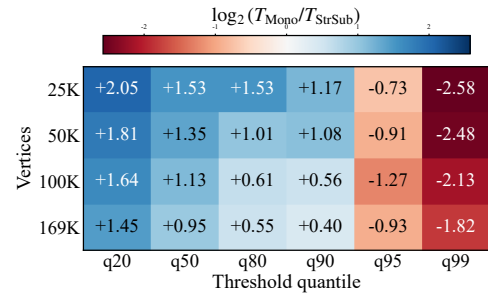


Figure 7: Relative runtime regimes remain stable across graph sizes.

6.5 RQ4: Parallel Scalability

Figure 8 reports FB15K-237 at q_{80} for STRSUB and ogbn-arxiv at q_{50} for MONOSEMMCE. Dynamic edge-root scheduling reduces MONOSEMMCE from 13.054 seconds on one thread to 2.323 seconds on 32

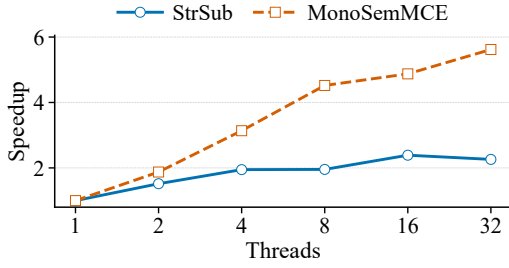


Figure 8: Parallel speedup under method-specific task scheduling.

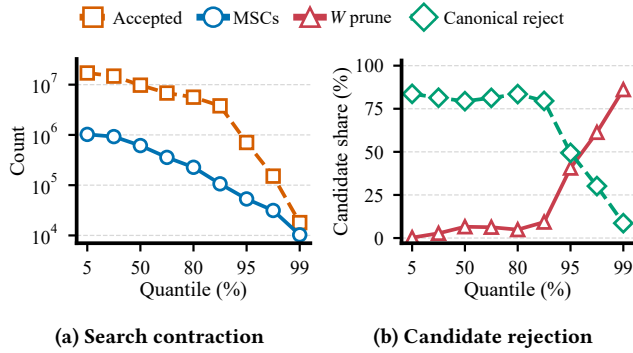


Figure 9: Canonical search contracts and shifts toward feasibility pruning.

threads, a $5.62\times$ speedup. The gain continues from 16 to 32 threads because canonical ownership exposes independent root subtrees without a shared visited table. Fixed-size subset partitioning reduces STRSUB from 19.662 seconds to 8.232 seconds at 16 threads, a $2.39\times$ speedup, and reaches $2.26\times$ at 32 threads. Separately collected phase counters attribute 98.88% of its 32-thread work to host-clique subset processing, while the final global filter takes only 65 ms. The saturation therefore arises inside the dominant search phase: structural maximal cliques induce tasks of unequal size, so a few large hosts determine the completion time even though equal-size subsets within each host are partitioned. The two curves consequently validate the parallelization choices described in Section 5: fine-grained canonical roots provide more uniform concurrency than host-clique decomposition.

6.6 RQ5: Search and Engineering Mechanisms

Figure 9 traces canonical-search diagnostics collected outside timing runs. Accepted states contract from 17.08 million at q_5 to 18,011 at q_{99} . At the same time, canonical rejection accounts for 83.6% of candidate tests at q_5 , whereas the W -based feasibility test prunes 86.1% at q_{99} . Thus high selectivity reduces work before child-state construction, matching the regime in which MONOSEMMCE becomes dominant. The ablations proceed hierarchically: they first isolate semantic pruning in STRSUB, then duplicate-state organization in MONOSEMMCE, and finally two method-specific engine primitives while holding the exact output fixed.

Component	Metric / workload	Full	Removed
Sorted-edge bound	tests (M) / nasa q_{99}	12.95	16.50
Vertex bound	tests (M) / nasa q_{99}	12.95	80.28
Local containment	GiB / FB15K q_{80}	0.12	2.26

Table 3: Effect of the three STRSUB pruning mechanisms.

Generation	ogbn q_{95}		nasa q_{80}	
	s	GiB	s	GiB
Canonical	0.388	0.39	0.729	0.22
Visited DFS	2.432	0.53	3.069	0.64
Layered	1.635	0.49	3.922	0.81

Table 4: Canonical ownership versus duplicate-state organizations.

Table 3 isolates the three STRSUB pruning mechanisms on their appropriate metrics. On sc-nasasrb at q_{99} , the sorted-edge bound reduces subset tests from 16.50 to 12.95 million, and the vertex bound reduces them from 80.28 to 12.95 million. Local containment is measured separately on FB15K q_{80} because its primary effect is peak memory rather than subset-test count, reducing 2.26 to 0.12 GiB.

The corresponding runtimes are 0.287 versus 0.324 seconds for the sorted-edge bound, 0.287 versus 0.312 seconds for the vertex bound, and 8.060 versus 11.685 seconds for local containment. Every variant retains the same 12,742 outputs, so the measured changes follow from pruning and local maximality checks while preserving enumeration semantics. Table 4 then holds the feasibility test, similarity cache, and final filter fixed while replacing only duplicate-state organization. *Visited DFS* suppresses duplicates with a global state table, whereas *Layered* materializes and deduplicates all feasible cliques of one size before generating the next. Canonical ownership is $4.2\times$ – $6.3\times$ faster and uses $1.3\times$ – $3.7\times$ less memory than these alternatives across the two workloads. Every variant returns the same normalized MSC set and size histogram, directly connecting the gain to unique-parent state generation rather than feasibility pruning.

Table 5 isolates the representation and state updates selected for each search. Each row changes one choice while preserving the remaining configuration; values are medians of three complete runs, and all variants return identical counts. Sparse lists fit SEMBK’s irregular candidate probes, while removing accumulator updates that its terminal-only test never reads eliminates redundant work. Blocked bitmaps and vector reordering support STRSUB’s repeated pivot-neighborhood operations. For MONOSEMMCE, incremental accumulation directly supplies the marginal contribution used by every child feasibility test. Complete results for all 31 evaluated configurations are archived with the artifact.

The paired speedups range from $1.24\times$ to $2.17\times$. More importantly, they follow the access pattern of each algorithm: sparse rows and direct sums avoid unused maintenance in SEMBK, bitmap intersections and reordering serve STRSUB’s repeated neighborhood operations, and accumulators serve the marginal tests performed at every MONOSEMMCE insertion. The ablation therefore supports

Method	Workload	Choice	With (s)	Without (s)	Speedup
SEMBK	Enron q_{80}	Sparse rows	8.106	10.572	1.30×
SEMBK	Enron q_{80}	Direct sum	10.572	22.935	2.17×
STRSUB	FB15K q_{80}	Bitmap	7.084	11.255	1.59×
STRSUB	FB15K q_{80}	Reorder	7.084	8.792	1.24×
MONOSEMCE	FB15K q_{99}	Accumulator	8.630	15.534	1.80×

Table 5: Paired engineering choices; other switches are fixed within each row, and speedup is without/with.

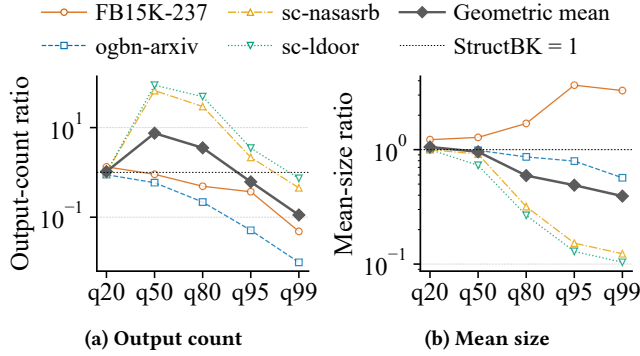


Figure 10: MSC output expands and then contracts as selectivity increases.

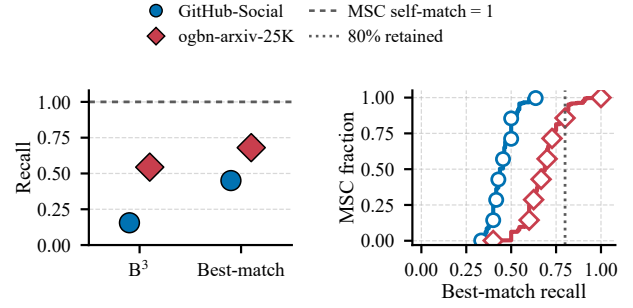
method-specific engine configurations while keeping the mathematical search procedures and outputs fixed.

6.7 RQ6: MSC Output Space

This experiment measures exact output counts and mean sizes on four graphs: FB15K-237, ogbn-arxiv, sc-nasasrb, and sc-lldoor. It uses five quantiles from q_{20} through q_{99} . Figure 10 normalizes each value by STRUCTBK on the same graph. The geometric-mean count ratios are 1.04, 7.50, 3.53, 0.61, and 0.11: intermediate thresholds split large structural cliques into many feasible subsets that are maximal by inclusion. Selective thresholds instead contract the result family. Mean size decreases from 1.06 to 0.39 of the structural reference over the same grid. The count peak and size decline describe the non-monotone search difficulty more precisely than either statistic alone: one structural host can yield many smaller MSCs in the intermediate regime, whereas high thresholds eliminate most extensions and leave fewer, smaller outputs. This expansion-contraction pattern explains both the intermediate STRSUB workload in RQ2 and the rapid contraction of the MONOSEMCE state space at selective thresholds in RQ5.

6.8 RQ7: Group Preservation and External Pattern Families

RQ7 first compares MSC with the every-pair definition. The reference cover contains all size-at-least-ten MSCs at q_{80} on GitHub-Social and the 25K induced ogbn-arxiv graph used in RQ3; the response cover contains every non-singleton PAIRSIM-MCE output at the same τ . For these overlapping covers, Extended B^3 Recall is the overlap-aware, entity-level recall of the reference cover and measures collective fragmentation [1]. Best-match recall $R_{\max}(C) =$



(a) Aggregate recovery (b) Per-group recovery
Figure 11: Pairwise thresholding fragments collectively feasible MSCs.

$\max_{P \in \mathcal{P}} |C \cap P|/|C|$ measures the largest fraction of MSC C retained by one output [17, 39].

Figure 11 shows the definition effect. PAIRSIM-MCE obtains B^3 Recall/best-match recall of 0.156/0.450 on GitHub-Social and 0.544/0.680 on ogbn-arxiv-25K. None of 605 GitHub-Social MSCs and only 0.32% of 624 ogbn-arxiv-25K MSCs is retained intact; respectively, 100% and 84.9% lose at least 20% of their members. The every-pair constraint therefore fragments groups that remain complete and collectively feasible.

A size-18 ogbn-arxiv MSC consists entirely of cs.IT papers on private information retrieval. Its average is 0.919 at $\tau = 0.889$, although 20 of 153 pairs fall below τ . Thus 13.1% of its within-group pairs violate the every-pair threshold while the complete group remains collectively coherent; this case instantiates the fragmentation measured by the cover-based metrics.

The external comparison fixes FastQC at $\gamma = 0.9$, FaPlex at $k = 2$, and uses STRUCTBK as the structural reference on the same q_{80} graphs. For two groups A and B , their match score is $F_1(A, B) = 2|A \cap B|/(|A| + |B|)$ [39]. Recovery averages each MSC’s best match in the baseline family, whereas fidelity averages each baseline result’s best match in the MSC family. This directional pair measures output-family agreement rather than ground-truth clustering.

Figure 12 separates the two matching directions. The left panel shows high MSC-to-baseline recovery on both datasets, whereas the right panel shows substantially lower baseline-to-MSC fidelity. MSC self-matching has score one by definition and appears only as the reference line. Recovery spans 0.851–0.992, whereas fidelity spans 0.123–0.269. The high recovery follows because every MSC is contained in some maximal clique, maximal quasi-clique, and maximal 2-plex; fidelity exposes the additional patterns introduced by those extensions. FastQC and FaPlex also differ from structural MCE: their mean induced edge densities are 0.942–0.952, and fewer than 0.05% of their outputs are cliques. Relative to the MSC family, their result unions cover $12.5\times$ – $27.6\times$ as many distinct vertices and have lower mean pairwise cosine similarity.

7 RELATED WORK

Clique Enumeration and Listing. BK backtracking [4], Tomita pivoting [36], and degeneracy ordering [11, 12] underpin exact

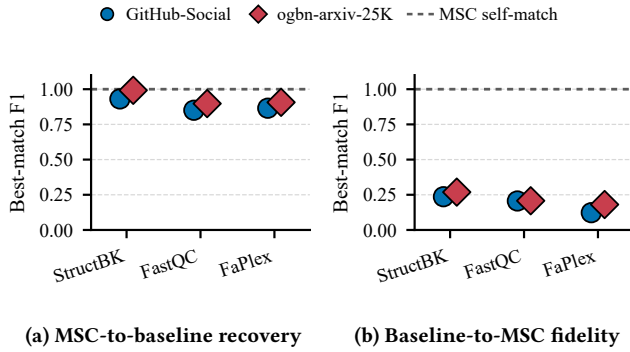


Figure 12: Structural and relaxed families achieve high MSC recovery but lower family fidelity.

MCE. The classical output bound and reporting perspective were established by Moon–Moser and subsequent maximal-clique analyses [5, 28]. Recent methods reduce BK subproblems [10, 38] or summarize clique families [26]; these methods preserve structural maximality and do not impose a group-level vector constraint. Large-graph implementations also exploit specialized search structures and parallel execution [6, 8, 25].

Constrained Cohesive Subgraphs. FastQC [42] and FaPlex [44] enumerate maximal quasi-cliques and k -plexes, respectively, while biplex methods relax complete bipartite structure [7]. They change the adjacency contract, whereas MSC retains a complete induced subgraph.

Earlier maximal similarity cliques threshold every pairwise similarity before applying structural MCE [14]; this is the contract represented by PAIRSIM-MCE in our evaluation. Average edge weight has also been used to select one high-scoring structural maximal clique for short-text segmentation [19]. Fairness-aware MCE constrains counts or balance among categorical vertex attributes [30]. The MSC contract enumerates all inclusion-wise maximal complete groups satisfying an aggregate threshold over real-valued vectors.

The closest constrained-enumeration model is maximal similar-biclique enumeration [41]: its hereditary constraint requires every vertex pair on one side to satisfy a similarity threshold. Similar-weight bicliques likewise constrain edge weights in a bipartite pattern [40]. MSC differs along both axes: it operates on cliques in a general graph and constrains the average of all pairwise vector similarities. This aggregate condition admits low-similarity pairs and is non-monotone under vertex insertion and deletion, which is precisely the source of the search problem studied here.

Attributed Community Search. Attributed-graph methods return partitions, such as CODICIL [33, 34, 43], or query-specific dense groups, such as attributed truss communities [15, 20]. Subspace and cross-graph variants likewise optimize partition- or query-oriented objectives [13, 16, 29]. These tasks do not enumerate inclusion-wise maximal complete groups under a vector aggregate, which gives MSC a distinct enumeration contract.

Vector Attributes. Representation models [3, 23, 31, 35] can supply the vectors. Given these fixed attributes, MSC evaluates the

aggregate threshold exactly; approximate vector retrieval addresses a separate nearest-neighbor task [2, 21, 27].

8 CONCLUSION

We formulated exact MSC enumeration under an average pairwise-similarity constraint and showed that its associated size decision problem is NP-complete. STRSUB combines structural decomposition with admissible semantic bounds; MONOSEMCE turns feasible minimum-contribution deletion into a canonical reverse-search forest. Their complementary threshold regimes jointly complete all 24 fixed dataset–threshold workloads, with MONOSEMCE $2.92\times$ – $6.24\times$ faster than STRSUB on every q_{99} row. Ablations connect these gains to layer pruning and canonical-state contraction, while natural-attribute results show when averaging preserves complete groups fragmented by pairwise thresholding. These results connect the group-level semantic definition to complementary exact search organizations.

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